

LAB #3 – Applications of Array Operations

Purpose:

Learn more on applications of mathematical operations on arrays with MATLAB. This is an individual lab.

Before the lab:

Read the course slides and chapter 3 of the textbook.

During the lab:

1. With MATLAB, finish problems 9 to 14 in the textbook (section 3.9).
2. In your textbook, look at the more advanced problems after number 20 (section 3.9), select 21, 23, 25 from those and finish them with MATLAB.
3. Close MATLAB.
4. **Tracing Exercises:** Evaluate what will be the values of variables in the table after these MATLAB instructions will be executed (**Do these by hand: Do not use MATLAB!**). Fill in the table below these questions with pencil and paper calculations. Must show all your calculation procedures.

```
n1 = [10 14 8 6]
n2 = [6 4 12 2]'
m = n1' + n2
n = n1 - n2
r = n1 .* n2
s = n1 .* n2'
```

```
n1 = [100:100:400]
p = n1' + n1'''
q = n1' + n1'
[t, u] = max (n1)
```

```
n1 = [10:20:60]
n2 = 'This is for fun?'
v = length (n2)
w = length (n1)
```

In your report, use a table as the following to display the results of step 4:

Variable Name	Variable Value
m	
n	
p	
q	
r	
s	
t	
u	
v	
w	

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Now complete the same three exercises using MATLAB. Are the results the same as expected? If not, please record the discrepancies and explain how MATLAB execute the code that way.

Lab Report Submission Requirement:

1. Submit Lab #3 based on the given templates on the Blackboard. Please post all the screenshots of the inputs and running outputs of each step on the lab manual.
2. Please submit your report on the blackboard by the due time.